

FINAL YEAR PROJECT PRESENTATION

Online Certificate Verification System

A Secure, Fast & Tamper-Proof Digital Verification Portal Built with Python, Flask & Cryptographic Hashing

Problem Statement

Challenges in Traditional Verification

- 1. Document Forgery & Tampering:** Physical degrees and certificates are easily duplicated or altered using modern editing tools.
- 2. Time-Consuming Process:** Manual verification by universities or background agencies takes days or even weeks.
- 3. High Operational Cost:** Processing physical verification requests incurs significant administrative expense and effort.

CYBER SECURITY TRAINING CERTIFICATE

 TechSecure Solutions

This is to certify that

JENNY HENRY

Successfully completed the **48 hour Advanced Cyber Security Training Course** with diligence, dedication and commitment.

Date of issue: 12.02.2025

Certificate NO: 536 7623 244



John Smith
Course instructor



Alice Bath
CEO

Proposed Solution



Cryptographic Security

Generates unique SHA-256 cryptographic hashes for every certificate to guarantee data integrity and eliminate fraud.



Instant QR Verification

Embeds a unique QR code on every document, enabling employers to verify authenticity in seconds with any camera.



Automated Portal

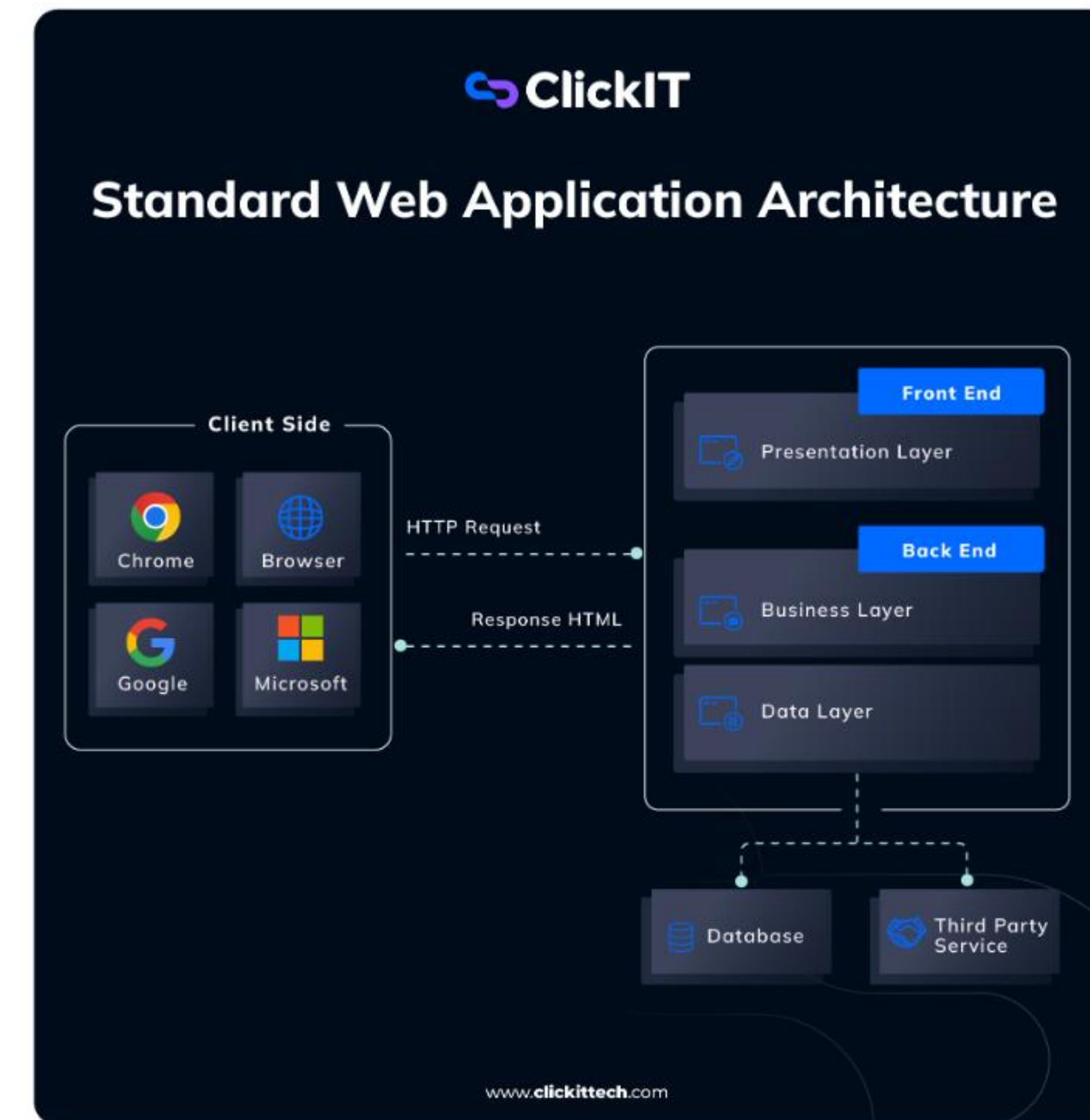
Provides a centralized web portal powered by Flask for seamless certificate generation, storage, and public lookups.

Core System Features

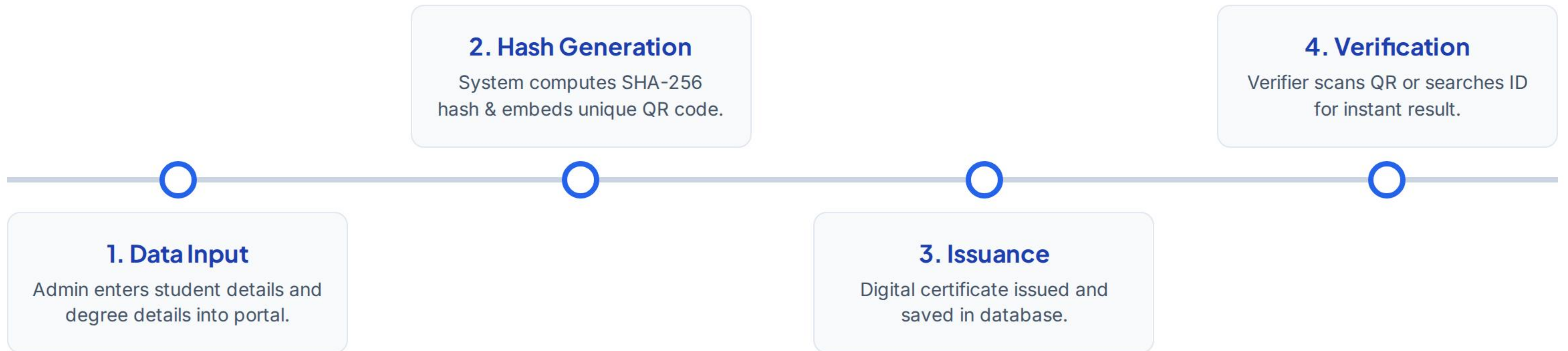
-  **Role-Based Access Control:** Distinct login workflows for Administrators (Issuers) and Public Users (Verifiers).
-  **Automated Certificate Issuance:** Direct generation of PDF certificates equipped with cryptographic hashes and embedded QR codes.
-  **Dual Verification Methods:** Verification via unique Certificate ID search or quick mobile QR code scanning.
-  **Centralized Relational Database:** Secure backend storage for student metadata, issue dates, and validation state.

System Architecture

-  **Backend:** Python with Flask micro-framework for lightweight, scalable routing and API operations.
-  **Frontend:** HTML5, CSS3, JavaScript, and Bootstrap for a clean, responsive user experience.
-  **Security Module:** Python hashlib for SHA-256 hash generation and integrity verification.
-  **Database:** SQLite / MySQL for structured storage of credentials and certificate records.



Workflow & Operations



Portal Capabilities

Admin / Issuer Module

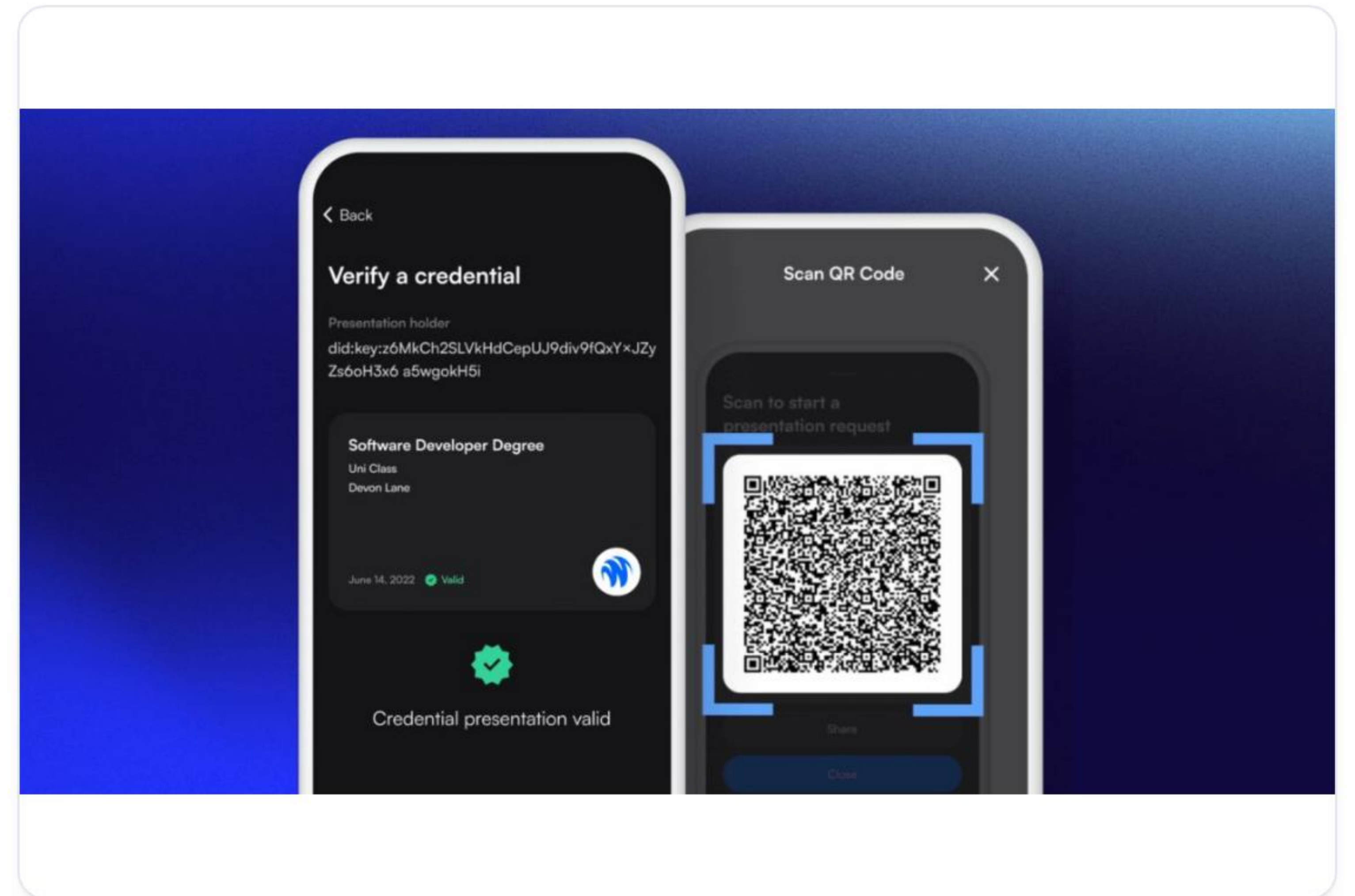
- Secure Authentication & Dashboard Access
- Single & Bulk Certificate Generation
- Certificate Revocation / Status Updates
- Audit Logging & Issuance History Tracking

Verifier / Public Portal

- No Registration Required for Verification
- Real-Time Validity Status Check
- Displays Verified Student Name & Major
- Instant Tamper Detection Notification

Instant QR Verification

- 📱 **Zero Friction Scanning:** Employers can scan the printed or digital QR code directly from any mobile camera.
- 🔗 **Direct Cryptographic Link:** The QR code embeds an encoded endpoint containing the certificate ID and cryptographic token.
- ✅ **Instant Feedback:** Displays official confirmation, issue date, and issuing institution details instantly.



System Impact Metrics

< 2s

Verification Time

Reduced from weeks to seconds

100%

Forgery Prevention

Protected by SHA-256 Hashing

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Paper Dependencies

Fully digital green workflow

System Comparison

Feature	Traditional Paper System	Proposed Online System
Verification Time	5 to 15 Business Days	Instant (< 2 Seconds)
Tamper Resistance	Low (Easy to fake physical stamps)	High (Cryptographic Hash Matching)
Verification Cost	High (Courier & Processing fees)	Near Zero (Fully Automated)
Accessibility	Manual In-person / Post	24/7 Global Web & QR Access

Future Scope

Blockchain Integration: Migrating to Ethereum Smart Contracts for decentralized immutability.

AI OCR Reader: Automatically reading physical certificate scans for legacy documents.

DigiLocker Integration: Direct API connection with government credential vaults.

Thank You!

Online Certificate Verification System project presentation is complete.

 Open for Questions & Feedback

DIGITAL
TECHNOLOGY

Image Sources



https://res.cloudinary.com/certifier/image/upload/v1739870788/security_training_certificate_templates_Certifier_blog_1_3549882727.webp

Source: certifier.io



<https://www.clickittech.com/wp-content/uploads/2024/04/Standard-Web-Application-Architecture.png>

Source: www.clickittech.com



https://cdn.prod.website-files.com/6311eb97e2519f5dbb9ea0cb/6439e3817c475b4b198752e1_3-digital%20certificates%20verifying%20a%20credential.jpg

Source: www.dock.io



https://img.magnific.com/premium-vector/abstract-dark-blue-digital-technology-background-with-futuristic-glowing-circular-wave-lines-cyan-light-curve-overlap-cyber-network-banner-ai-science-global-big-data-business-presentation_258787-8096.jpg?semt=ais_hybrid&w=740&q=80

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